

DECT-2020 NR: A Comprehensive Systematic Literature Review, Taxonomy, and Future Research Directions

Ashuqullah Alizai^{a,*}, Mohammad Reza Mousavi^a, Stephan Ludwig^a

^aAalen University of Applied Sciences, Aalen, Germany

Abstract

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1. Introduction

1.1. Background and Motivation

The rapid digital transformation of modern industry has fundamentally changed the role of wireless communication systems. Emerging paradigms such as Industry 4.0, the Industrial Internet of Things (Industrial Internet of Things (IIoT)), cyber-physical systems (Cyber-Physical Systems (CPS)), intelligent manufacturing, autonomous robotics, digital twins, and edge computing increasingly rely on seamless connectivity among heterogeneous devices, sensors, actuators, and control systems. Unlike traditional communication networks that primarily support human-centric services, modern industrial environments require continuous machine-to-machine communication capable of supporting real-time monitoring, distributed control, predictive maintenance, collaborative robotics, and autonomous decision-making. Consequently, wireless communication has evolved from a convenient alternative to wired connectivity into a fundamental enabling technology for digital transformation across manufacturing, transportation, healthcare, logistics, agriculture, and other industrial sectors [1, 2, 3, 4].

The communication requirements of these emerging applications differ significantly from those of conventional mobile broadband services. Industrial communication systems demand deterministic medium access, ultra-high reliability, low end-to-end latency, synchronized operation among distributed devices, massive machine connectivity, high energy efficiency, and predictable communication performance even under harsh electromagnetic conditions. Simultaneously, wireless infrastructures must support large numbers of heterogeneous devices while maintaining scalability, deployment flexibility, and long operational life-

times with minimal maintenance. These often conflicting requirements have become one of the principal driving forces behind the evolution of next-generation wireless communication technologies [5, 2, 1].

Although existing wireless technologies such as Wi-Fi, Bluetooth, IEEE 802.15.4, WirelessHART, and conventional cellular systems have enabled numerous industrial applications, each has been designed to optimize only a subset of these communication requirements. High-throughput technologies frequently struggle to provide deterministic communication and predictable latency, whereas low-power wireless networks often sacrifice scalability, mobility, or spectral efficiency. Similarly, traditional cellular systems provide extensive coverage and mobility support but generally rely on licensed spectrum and infrastructure-centric deployments that may not be suitable for all industrial scenarios. These limitations have motivated the development of new radio access technologies capable of combining the flexibility of license-exempt operation with the reliability, scalability, and quality-of-service required by next-generation industrial communication systems [4, 6, 7].

1.2. Evolution toward IMT-2020 Wireless Communications

The evolution of wireless communication has been characterized by continuous improvements in capacity, spectral efficiency, mobility support, and service diversity. While second- and third-generation cellular systems primarily focused on voice and mobile data services, fourth-generation technologies introduced all-IP architectures capable of supporting broadband multimedia applications. However, the rapid emergence of industrial automation, cyber-physical systems, and large-scale machine communications exposed new requirements that could not be fully addressed through incremental improvements of existing cellular technologies. Future wireless networks were expected to simultaneously support enhanced broadband services, latency-critical industrial applications, and

*Corresponding author.

Email addresses: ashuqullah.alizai@hs-aalen.de (Ashuqullah Alizai), mohammad.mousavi@hs-aalen.de (Mohammad Reza Mousavi), stephan.ludwig@hs-aalen.de (Stephan Ludwig)

massive deployments of connected devices, creating the need for a fundamentally new communication framework [8, 6, 9, 7, 10, 11].

To address these challenges, the International Telecommunication Union Radiocommunication Sector (ITU-R) established the International Mobile Telecommunications-2020 (IMT-2020) framework, defining a comprehensive vision for future wireless communication systems. Rather than specifying a single radio technology, IMT-2020 establishes performance objectives, minimum technical requirements, evaluation methodologies, and interoperability criteria that candidate radio interface technologies must satisfy. The framework defines three complementary service categories, namely enhanced Mobile Broadband (Enhanced Mobile Broadband (eMBB)), Ultra-Reliable Low-Latency Communications (Ultra-Reliable Low-Latency Communications (URLLC)), and massive Machine-Type Communications (Massive Machine-Type Communications (mMTC)), which together extend wireless communication far beyond conventional mobile broadband toward industrial automation, mission-critical networking, intelligent transportation, and massive IoT deployments [12, 13, 14, 15].

An important characteristic of IMT-2020 is its technology-neutral philosophy. Unlike previous generations, which were largely associated with a single family of cellular technologies, IMT-2020 establishes a common performance, evaluation, and interoperability framework that enables multiple Radio Interface Technologies (RITs) to satisfy the same set of technical requirements rather than prescribing a single air interface [12, 13, 14, 15]. Although 3GPP New Radio (NR) represents the dominant cellular implementation within the IMT-2020 ecosystem, the framework also accommodates alternative radio technologies designed for deployment scenarios where conventional cellular solutions may not provide the most appropriate balance between performance, deployment flexibility, and spectrum utilization.

Within this context, European Telecommunications Standards Institute (ETSI) initiated the development of DECT-2020 New Radio (DECT-2020 NR) as a next-generation wireless technology operating in globally available license-exempt spectrum. Building upon the legacy of the original Digital Enhanced Cordless Telecommunications (DECT) standard while introducing a completely new radio interface, DECT-2020 NR was designed to support reliable, scalable, and deterministic wireless communication for industrial automation, massive machine-type communications, professional audio, and other mission-critical applications. Following a comprehensive evaluation against the technical performance requirements defined by IMT-2020, DECT-2020 NR was officially approved by the International Telecommunication Union Radiocommunication Sector (ITU-R) as a RIT, becoming the first ETSI-developed non-cellular radio interface included in the IMT-2020 family [16, 17, 18, 19, 20]. This achieve-

ment significantly broadened the technological diversity of IMT-2020 by introducing a wireless communication platform specifically optimized for industrial, enterprise, and other license-exempt deployment scenarios, thereby complementing existing cellular solutions rather than replacing them [21, 22].

1.3. DECT-2020 NR: A New IMT-2020 Radio Interface

Among the radio interface technologies accepted within the IMT-2020 framework, DECT-2020 New Radio (DECT-2020 NR) represents a unique milestone in the evolution of wireless communications. Unlike conventional cellular technologies, DECT-2020 NR was specifically developed by the European Telecommunications Standards Institute (ETSI) to provide a flexible, scalable, and reliable wireless communication system operating in globally available license-exempt spectrum. Building upon the successful legacy of Digital Enhanced Cordless Telecommunications (DECT), the new radio interface extends the technology far beyond cordless telephony by introducing a modern packet-oriented architecture capable of supporting industrial automation, massive machine-type communications, ultra-reliable low-latency services, and mission-critical wireless networking. Its acceptance by ITU-R as an official IMT-2020 radio interface established DECT-2020 NR as the first non-cellular technology recognized within the IMT-2020 family, significantly broadening the technological diversity of future wireless communication systems [15, 16, 18, 19].

The standardization of DECT-2020 NR was carried out through a comprehensive family of ETSI Technical Specifications defining the overall system architecture, protocol stack, physical layer, medium access control procedures, convergence layer, and implementation framework. Release 1 established the initial protocol architecture and radio interface, while Release 2 introduced numerous enhancements aimed at improving system flexibility, scalability, interoperability, and industrial applicability. Together with the accompanying ETSI Technical Reports, these specifications provide a complete framework covering protocol operation, spectrum usage, regulatory aspects, co-existence mechanisms, and evaluation procedures required for IMT-2020 compliance [23, 24, 25, 26, 27, 17, 16].

Unlike conventional cellular deployments that depend on licensed spectrum and centralized network infrastructures, DECT-2020 NR emphasizes autonomous network operation, decentralized resource management, efficient spectrum utilization, and flexible deployment in industrial and enterprise environments. These characteristics make the technology particularly attractive for private wireless networks, industrial Internet of Things applications, cyber-physical production systems, warehouse automation, autonomous mobile robots, smart logistics, and other mission-critical industrial scenarios where deterministic communication, low deployment cost, and spectrum independence are essential. Consequently, DECT-2020 NR has attracted increasing attention from both academia

and industry as a complementary wireless technology capable of addressing communication scenarios that remain challenging for conventional cellular systems [20, 19, 21].

1.4. Evolution of the DECT-2020 NR Research Landscape

Following the publication of the ETSI specifications, research on DECT-2020 NR rapidly evolved from standardization activities toward practical implementation and experimental validation. Early studies concentrated on validating the protocol architecture and investigating the fundamental characteristics of the physical and medium access control layers, including synchronization procedures, waveform generation, channel estimation, modulation and coding schemes, random access mechanisms, coexistence strategies, and radio resource management. These investigations confirmed the feasibility of the standardized design while establishing the theoretical and experimental foundations required for practical deployment [21, 22, 28, 29].

As implementation platforms became available, the focus of the research community progressively shifted toward hardware realization and system-level experimentation. Software-defined radio platforms, FPGA-based implementations, commercial Nordic Semiconductor development kits, embedded communication modules, and industrial testbeds enabled comprehensive experimental investigations under realistic deployment conditions. These studies evaluated protocol behavior, link performance, latency, throughput, synchronization accuracy, coexistence performance, and energy consumption, thereby demonstrating the practical viability of DECT-2020 NR for industrial wireless communication systems [30, 31, 32, 33, 34].

More recently, the research landscape has expanded considerably beyond the lower communication layers. Current investigations encompass scheduling and radio resource allocation, routing and multi-hop communication, mesh networking, positioning and localization, authentication and security, coexistence with heterogeneous wireless technologies, software-defined networking, energy-efficient communication, multi-RAT integration, industrial automation, professional audio applications, and emerging non-terrestrial communication scenarios. This diversification illustrates the rapid transition of DECT-2020 NR from a newly standardized radio interface into a comprehensive research field spanning wireless communications, embedded systems, industrial networking, and cyber-physical systems [35, 36, 37, 38, 39, 40, 41, 42, 43].

1.5. Research Gap

The rapid evolution of DECT-2020 NR has resulted in a substantial increase in both the volume and diversity of the available scientific literature. As demonstrated by the systematic review presented in this paper, research has progressively evolved from standardization activities toward protocol optimization, hardware implementation, experimental validation, and industrial deployment. Consequently, the literature now encompasses a

broad spectrum of research topics, including system architecture, Physical Layer (PHY) techniques, Medium Access Control (MAC) protocols, synchronization, scheduling and resource allocation, routing and mesh networking, localization and positioning, security, coexistence mechanisms, embedded implementations, performance evaluation, and application-specific deployments. This diversification demonstrates the growing technological maturity of DECT-2020 NR while simultaneously increasing the complexity of understanding the overall research landscape [23, 24, 25, 26, 27, 21, 22, 28, 31, 32, 34, 35, 37, 41].

The findings of our systematic literature review further indicate that the available knowledge is distributed across multiple publication types and research communities. Fundamental aspects of the technology are documented in ETSI Technical Specifications, Technical Reports, and ITU-R recommendations, whereas implementation methodologies, protocol optimizations, hardware platforms, experimental evaluations, and industrial applications are primarily reported in journal articles, conference proceedings, industrial demonstrations, and experimental studies. Although these publications collectively provide comprehensive coverage of individual research topics, they are largely organized according to specific technical objectives rather than the overall evolution of the technology. Consequently, researchers and practitioners must consult numerous heterogeneous sources to obtain a comprehensive understanding of DECT-2020 NR and its current state of development [12, 15, 16, 17, 23, 24, 25, 26, 27, 21, 19, 22, 31, 33].

Furthermore, the continuous evolution of the standard itself has accelerated the diversification of research. Following the publication of the initial Release 1 specifications, Release 2 introduced additional capabilities that stimulated new investigations in radio resource management, synchronization, localization, industrial automation, energy-efficient communication, multi-hop networking, and emerging Industry 4.0 applications. At the same time, the availability of commercial chipsets, software development kits, Field-Programmable Gate Array (FPGA) implementations, and industrial testbeds has shifted the research focus from theoretical protocol design toward comprehensive experimental validation under realistic deployment conditions. As a result, the literature has become increasingly fragmented across protocol layers, implementation platforms, optimization techniques, and application domains [23, 24, 25, 30, 22, 31, 38, 40, 42, 43].

Our analysis of the selected literature further reveals that most existing publications investigate individual aspects of DECT-2020 NR in isolation. While these studies have significantly advanced the technology, relatively few establish relationships among different protocol layers, compare alternative technical approaches across multiple research domains, or synthesize the overall evolution of the technology from standardization to industrial deployment. Consequently, common design principles, emerging research trends, remaining technical challenges, and fu-

ture research opportunities remain dispersed throughout the literature rather than being presented within a unified analytical framework [21, 22, 35, 37, 41, 31].

Motivated by these observations, this paper presents a comprehensive systematic literature review of DECT-2020 NR following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 methodology. Unlike conventional narrative surveys, our review employs a transparent and reproducible protocol for literature identification, screening, quality assessment, data extraction, and qualitative synthesis. The selected publications are systematically classified into a unified taxonomy covering the complete DECT-2020 NR technology stack, including standardization, system architecture, PHY, MAC, scheduling and resource allocation, routing and mesh networking, localization, security, implementation platforms, experimental validation, performance evaluation, and industrial applications. By integrating these diverse research directions into a coherent analytical framework, this survey provides a comprehensive overview of the current state of DECT-2020 NR research while identifying technological trends, remaining research challenges, and promising future research directions.

1.6. Contributions of This Survey

The main contributions of this survey are summarized as follows:

1. We present the first comprehensive PRISMA-based systematic literature review of DECT-2020 NR research, employing a transparent, reproducible, and evidence-based methodology for literature identification, screening, quality assessment, data extraction, and qualitative synthesis.
2. We provide a unified taxonomy that systematically classifies DECT-2020 NR research into coherent technical categories spanning the complete technology stack, including standardization, system architecture, physical layer, medium access control, scheduling and resource allocation, routing and mesh networking, localization, security, implementation platforms, experimental validation, performance evaluation, and industrial applications.
3. We critically synthesize the existing literature by integrating ETSI standards, ITU-R recommendations, journal articles, conference papers, and industrial implementation studies, thereby providing a comprehensive overview of the technological evolution of DECT-2020 NR from standardization to practical deployment.
4. We analyze the current state of research across the identified technical categories, highlighting the major technological developments, implementation approaches, experimental achievements, and emerging research trends that characterize the evolution of DECT-2020 NR.

5. We identify the remaining research challenges, unresolved technical issues, and promising future research directions, providing guidance for researchers, system developers, standardization bodies, and industrial practitioners working on next-generation industrial wireless communication systems.

1.7. Paper Organization

The remainder of this paper is organized as follows. Section 2 presents the systematic review methodology, including the search strategy, study selection, quality assessment, data extraction, and qualitative synthesis procedures. Section 3 summarizes the literature search results, publication trends, and bibliometric analysis. Section ?? provides a comprehensive technical synthesis of the selected studies according to the proposed taxonomy. Section 7 discusses the remaining research challenges and outlines promising future research directions. Finally, Section 8 concludes the paper.

2. Methodology

This study adopts a Systematic Literature Review (SLR) methodology following the PRISMA 2020 guidelines to provide a transparent, systematic, and reproducible review of the existing literature on DECT-2020 NR. Unlike conventional narrative surveys, the PRISMA framework employs a predefined review protocol that minimizes selection bias through a structured process comprising literature identification, screening, eligibility assessment, quality assessment, data extraction, and qualitative synthesis.

The review protocol was established prior to the literature search to ensure methodological consistency and reproducibility throughout the review process. As illustrated in Figure ??, the review consisted of nine sequential stages:

1. Formulation of the Research Questions (RQs);
2. Selection of digital literature databases;
3. Development of the search strategy;
4. Systematic literature retrieval;
5. Duplicate detection and removal;
6. Study selection using predefined inclusion and exclusion criteria;
7. Quality Assessment (QA) of the selected studies;
8. Structured data extraction;
9. Qualitative and comparative data synthesis.

2.1. Review Protocol

A review protocol was defined before commencing the literature search to ensure the transparency, consistency, and reproducibility of the entire review process. The protocol specifies the research objectives, search strategy, study selection criteria, quality assessment procedure, data extraction framework, and synthesis methodology

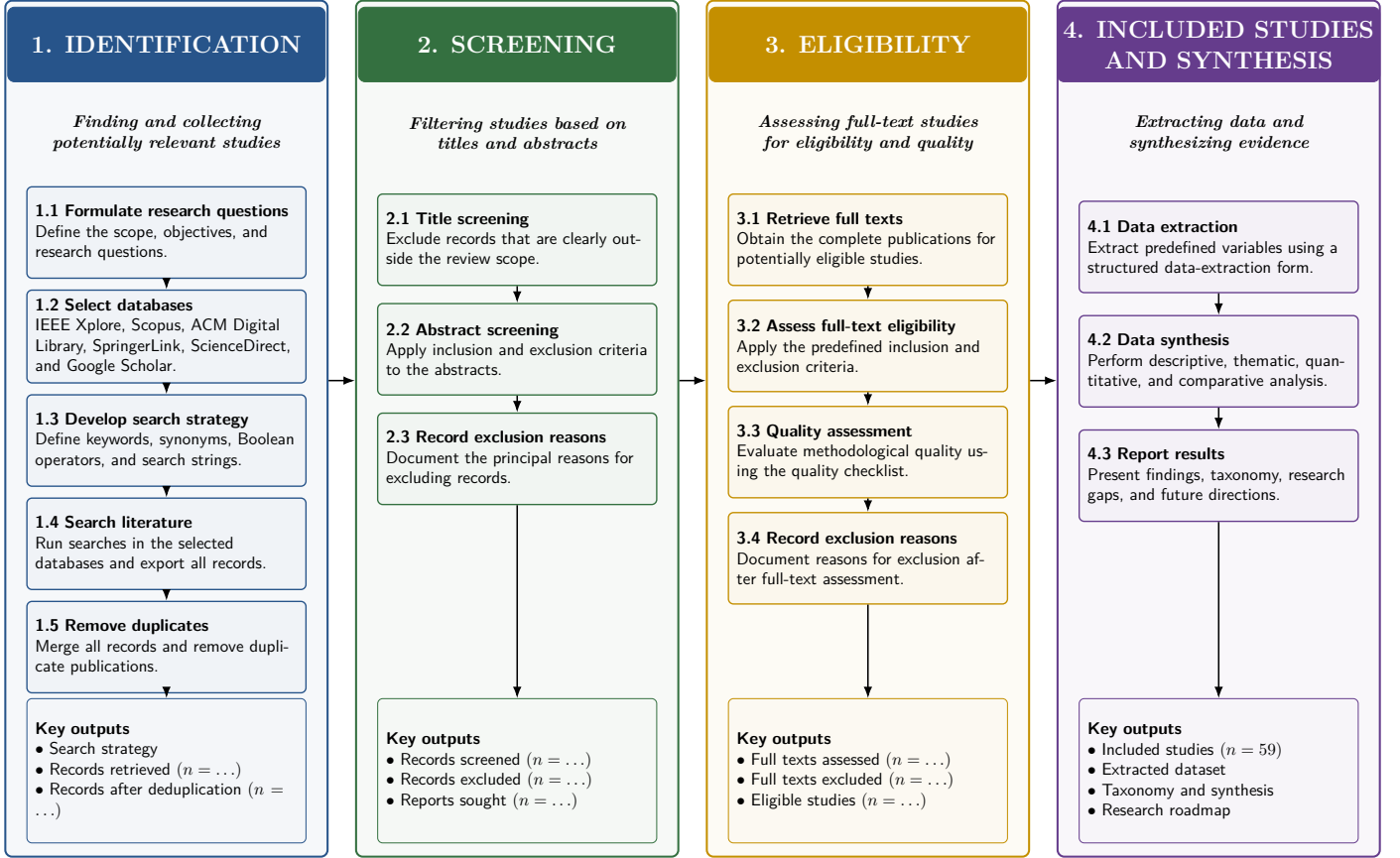


Figure 1: PRISMA-oriented systematic review workflow used in this study. The first three phases describe study identification, screening, and eligibility assessment, while the final phase covers extraction, synthesis, and reporting of the included evidence.

adopted in this study. Although the protocol was not formally registered in an external repository, all methodological decisions were established before the study selection process began.

Table 1 summarizes the review protocol adopted in this work.

2.2. Research Questions

The primary objective of this SLR is to provide a comprehensive overview of the current state of DECT-2020 NR research, identify emerging research trends, evaluate existing implementation and validation approaches, and highlight future research opportunities. To achieve these objectives, the review addresses the following RQs.

- RQ1** How has research on DECT-2020 NR evolved in terms of publication trends, publication types, publishers, and application domains?
- RQ2** Which technical aspects of DECT-2020 NR, including the PHY, MAC, routing, resource allocation, positioning, security, and industrial applications, have been investigated?
- RQ3** Which evaluation methodologies, implementation platforms, software tools, and hardware platforms

Table 1: Summary of the Review Protocol

Item	Description
Review Method	PRISMA 2020 SLR
Research Domain	DECT-2020 NR
Publication Period	January 2020 – July 2026
Language	English
Primary Databases	IEEE Xplore, Scopus, ACM Digital Library, SpringerLink, ScienceDirect
Complementary Search	Google Scholar (Backward and Forward Snowballing)
Search Fields	Title, Abstract, and Author Keywords
Duplicate Management	Zotero Reference Manager
Study Selection	PRISMA Screening Process
Quality Assessment	Five quality criteria (0–10)
Data Extraction	Structured extraction template
Data Analysis	Descriptive, comparative, and thematic synthesis

have been employed to validate DECT-2020 NR systems?

RQ4 Which Key Performance Indicators (KPIs) have been used to evaluate DECT-2020 NR, and what performance characteristics have been reported?

RQ5 What are the principal scientific contributions and technical limitations reported in the existing literature?

RQ6 Which research gaps remain insufficiently addressed in the current body of DECT-2020 NR literature?

RQ7 What future research directions are most promising for advancing DECT-2020 NR towards large-scale IIoT deployments?

2.3. Eligibility Criteria

The inclusion and exclusion criteria were defined before the screening process. A study was included if it satisfied all of the following criteria:

- it primarily investigated DECT-2020 NR or DECT NR+;
- it presented a technical, experimental, analytical, architectural, or application-oriented contribution;
- it was written in English;
- the full text was available;
- it was published between January 2020 and July 2026;
- it provided sufficient methodological or technical detail to support data extraction.

A study was excluded if it:

- mentioned DECT-2020 NR only briefly or as background;
- primarily addressed another wireless technology;
- was a duplicate record;
- lacked sufficient technical content;
- was not available in English;
- was an editorial, presentation slide set, poster, or non-technical document.

Studies that were relevant to the broader context of industrial wireless communications but did not primarily focus on DECT-2020 NR were classified as contextual studies and were excluded from the quantitative synthesis.

2.4. Quality Assessment

Each included study was assessed using five quality criteria:

1. clarity of the research objective;
2. appropriateness of the methodology;
3. strength of the validation;
4. clarity of the discussion and limitations;
5. reproducibility of the study.

Each criterion was scored on a three-point scale:

- 0: criterion not addressed;
- 1: criterion partially addressed;
- 2: criterion fully addressed.

The total quality score therefore ranged from 0 to 10. Studies scoring 8–10 were classified as high quality, studies scoring 5–7 as moderate quality, and studies scoring below 5 as low quality.

2.5. Data Extraction

A structured data extraction form was used to collect information from each included study. The extracted fields comprised:

- bibliographic information;
- publication year and publication type;
- primary and secondary research contributions;
- technical topics;
- evaluation methodology;
- hardware and software platforms;
- network topology and deployment environment;
- frequency band and bandwidth;
- evaluated KPIs;
- application domain;
- key findings;
- scientific contribution;
- limitations;
- identified research gaps;
- proposed future work.

The extracted information was stored in a structured spreadsheet database. Binary one-hot coding was used for multi-label attributes such as technical topics, evaluation methods, application domains, and performance metrics.

2.6. Study Classification and Coding

Each study was assigned exactly one primary contribution category, while multiple secondary technical topics were allowed. This distinction prevented double counting in the analysis of primary research contributions while preserving the multidisciplinary nature of the literature.

Multi-label technical topics included the PHY, MAC, routing and mesh networking, scheduling and resource allocation, positioning, security, energy efficiency, coexistence, artificial intelligence, standardization, industrial applications, URLLC, and mMTC.

Evaluation methodologies were coded independently as simulation, analytical or mathematical modeling, experimental or testbed validation, and survey or overview. A paper could therefore belong to more than one methodological category.

2.7. Reliability and Consistency Checks

To improve coding reliability, the extracted data were reviewed in multiple stages. Evaluation methods, technical topics, performance metrics, hardware platforms, and application domains were cross-checked across the master database and the corresponding binary classification matrices.

Where inconsistencies were identified, the full paper, abstract, methodology, results, and conclusion sections were re-examined. The final database was then frozen and used as the single source for all quantitative analyses, tables, figures, and percentages reported in this review.

2.8. Data Synthesis and Analysis

The included studies were analyzed using descriptive, comparative, and thematic synthesis.

Descriptive analysis was used to examine publication trends, publication types, research categories, evaluation methods, hardware platforms, application domains, and performance metrics.

Comparative analysis was used to examine differences between simulation-based, analytical, and experimental studies.

Thematic synthesis was used to group the reported scientific contributions, limitations, research gaps, and future research directions into recurring themes.

Because technical topics, evaluation methods, and performance metrics were coded as multi-label attributes, the corresponding percentages may sum to more than 100%.

2.9. Threats to Validity

Several threats to validity should be acknowledged. First, relevant studies may have been missed because of differences in database indexing or search terminology. This risk was reduced by searching multiple databases and performing backward and forward snowballing.

Second, the classification of research topics and contributions involves researcher judgment. To reduce subjectivity, a predefined coding framework was applied consistently, and ambiguous cases were re-examined using the full text.

Third, the rapidly evolving nature of DECT-2020 NR means that recently accepted or presented papers may not yet be indexed in major databases. Therefore, the review reflects the state of the literature up to July 2026.

Finally, some studies did not provide complete information about hardware, software, datasets, or reproducibility, which limited the depth of certain comparisons.

3. Results of the Systematic Literature Review

This section presents the results of the systematic literature review conducted according to the methodology described in Section 2. The objective of this section is to quantitatively characterize the collected literature rather than to discuss the technical contributions of individual publications. Accordingly, the analysis focuses on the outcome of the literature search, bibliometric characteristics of the selected studies, research trends, technical topic distribution, evaluation methodologies, validation approaches, reported performance metrics, and implementation platforms. Together, these analyses provide a comprehensive overview of the current DECT-2020 NR research landscape and establish the foundation for the technical synthesis presented in Section ??.

3.1. Literature Search Results

The systematic literature search was performed in July 2026 following the review protocol presented in Section 2. Five major scientific databases, namely IEEE Xplore, Scopus, ACM Digital Library, SpringerLink, and ScienceDirect, were systematically searched using the predefined Boolean search query. In addition, Google Scholar was employed as a complementary search engine to support backward and forward snowballing and to identify relevant publications that were not indexed in the primary scientific databases. The complete study selection process followed the PRISMA 2020 framework and is illustrated in Figure ??.

The initial database search identified 86 potentially relevant publications, comprising 42 publications retrieved from IEEE Xplore and 44 publications indexed in Scopus. No additional publications satisfying the predefined search criteria were identified in ACM Digital Library, SpringerLink, or ScienceDirect. Google Scholar returned a substantially larger number of search results; however, these were not considered part of the primary search dataset and were instead used exclusively for citation tracking and snowballing.

After removing duplicate records using Zotero Reference Manager, the remaining publications were screened according to the predefined inclusion and exclusion criteria.

The screening process consisted of title screening, abstract screening, and full-text assessment. Publications unrelated to DECT-2020 NR, papers lacking sufficient technical content, inaccessible documents, and studies outside the scope of this review were excluded. Additional relevant publications identified through backward and forward snowballing were incorporated into the final literature corpus whenever they satisfied all eligibility criteria and were not already included in the primary database search.

The final dataset comprises 51 high-quality publications together with the relevant ETSI Technical Specifications, ETSI Technical Reports, and ITU-R recommendations that define the DECT-2020 NR standardization framework. Collectively, these documents capture the evolution of DECT-2020 NR from its initial standardization activities through protocol design, algorithm development, hardware implementation, experimental validation, and industrial deployment. This curated dataset forms the basis for the quantitative analyses presented in the following subsections and for the comprehensive technical synthesis provided in Section ??.

3.2. Bibliometric Analysis

To obtain a comprehensive understanding of the evolution of DECT-2020 NR research, the final literature corpus was analyzed from multiple bibliometric perspectives. Unlike the subsequent technical synthesis, which examines the technical contributions of the literature, the objective of this section is to characterize the research landscape quantitatively. The analysis investigates publication trends over time, publication types, primary research contributions, technical topic distribution, evaluation methodologies, validation approaches, reported performance metrics, and implementation platforms. These statistics provide valuable insights into the maturity of the research field, the dominant research directions, and the methodological evolution of the published literature.

3.2.1. Publication Trends

Figure ?? illustrates the annual distribution of the publications included in this systematic literature review. The figure clearly demonstrates the rapid growth of DECT-2020 NR research since the introduction of the technology. Following the publication of the earliest standardization activities in 2020, the number of scientific publications increased steadily during the subsequent years, reflecting the growing academic and industrial interest in the technology.

The publication trend indicates three distinct phases in the evolution of the research field. The first phase (2020–2022) was characterized primarily by standardization activities, architectural design, and the establishment of the fundamental communication framework. During this period, research focused on defining the protocol architecture, physical layer, medium access control procedures, and demonstrating compliance with the IMT-2020 requirements. The second phase (2023) represents a transition toward implementation-oriented research, during which the

availability of development platforms enabled the first experimental investigations and practical demonstrations. The third phase (2024–2026) exhibits a substantial increase in research activity, with publications expanding toward protocol optimization, localization, routing, scheduling, industrial applications, hardware implementations, and large-scale experimental evaluations.

The apparent reduction in the number of publications during 2026 should not be interpreted as a decline in research activity. Instead, it reflects the fact that the literature search was conducted during the middle of the publication year, and therefore many conference proceedings and journal articles scheduled for publication later in the year were not yet available. Overall, the observed publication trend demonstrates that DECT-2020 NR has rapidly evolved from a newly standardized radio interface into an active and expanding research field attracting increasing attention from both academia and industry.

3.2.2. Publication Types

The distribution of publication types is presented in Figure ?. Conference papers constitute the dominant publication venue, accounting for the majority of the included studies, whereas journal articles, survey papers, and academic theses represent comparatively smaller proportions of the literature.

This distribution reflects the relatively recent emergence of DECT-2020 NR as an IMT-2020 radio interface. Newly developing wireless communication technologies are typically disseminated first through international conferences, where researchers rapidly communicate novel algorithms, protocol enhancements, hardware implementations, and experimental results. Journal publications generally appear later as the technology matures and more comprehensive validation studies become available. Consequently, the predominance of conference publications observed in this review is consistent with the current developmental stage of the technology.

The presence of several survey papers further indicates that the research community has begun consolidating knowledge across individual technical domains. Nevertheless, the relatively limited number of comprehensive surveys demonstrates that the research landscape is still evolving, thereby motivating the need for a systematic literature review that integrates standardization documents, protocol developments, implementation studies, and experimental evaluations into a unified analytical framework.

3.3. Research Characteristics

Beyond the bibliometric evolution of the literature, the selected publications were further analyzed according to their primary research contributions, technical focus areas, evaluation methodologies, validation approaches, reported performance metrics, and implementation platforms. Unlike conventional bibliometric indicators, these characteristics provide insight into how the DECT-2020 NR research

community has progressed from standardization activities toward algorithm development, implementation, experimental validation, and industrial deployment. Collectively, these analyses reveal the maturity of the research field and identify the dominant methodological trends that have shaped the evolution of DECT-2020 NR.

3.3.1. Primary Research Contributions

Figure 4 classifies each publication according to its primary scientific contribution. The results indicate that the current research landscape is dominated by industrial applications, physical-layer techniques, and medium access control mechanisms, which together represent the largest proportion of the available literature. These research areas constitute the technological foundation of DECT-2020 NR and therefore received considerable attention during the early stages of the technology’s development.

A substantial number of publications also focus on standardization activities, routing and mesh networking, and analytical modeling, reflecting the continued evolution of the protocol stack beyond the initial ETSI specifications. In contrast, comparatively fewer studies investigate scheduling and resource allocation, localization, security, and multi-radio interoperability. Although these research topics have attracted increasing interest in recent years, they remain comparatively underexplored relative to the core communication protocols. This distribution suggests that future research is expected to expand toward higher-layer optimization, intelligent resource management, and cross-layer communication techniques as the technology continues to mature.

3.3.2. Technical Topic Coverage

The multi-label classification presented in Figure 5 provides a broader perspective on the technical coverage of the reviewed literature. Unlike the primary contribution analysis, individual publications may contribute to multiple technical domains simultaneously, thereby illustrating the interdisciplinary nature of DECT-2020 NR research.

The analysis demonstrates that physical-layer techniques, medium access control, routing and mesh networking, and experimental applications constitute the most extensively investigated research areas. Considerable research activity is also observed in standardization, URLLC, energy-efficient communication, and positioning technologies, highlighting the growing diversification of the research landscape. Conversely, comparatively limited attention has been devoted to queue management, artificial intelligence-assisted networking, backscatter communication, and advanced resource allocation mechanisms. These findings indicate that while the fundamental communication architecture has reached a relatively mature stage, several higher-layer optimization topics remain open research directions.

3.3.3. Evaluation Methodologies

Figure 6 summarizes the evaluation methodologies adopted by the reviewed publications. Simulation represents the dominant evaluation approach, followed by analytical modeling and experimental validation. Survey papers constitute a comparatively smaller proportion of the literature, reflecting the relatively recent emergence of DECT-2020 NR research.

The predominance of simulation-based studies is consistent with the early development phase of a new wireless communication technology, where protocol verification, algorithm evaluation, and system optimization are typically performed before large-scale hardware implementation becomes widely available. Nevertheless, the significant number of experimental studies demonstrates that the research community has increasingly shifted toward practical validation using prototype hardware, software-defined radio platforms, commercial devices, and industrial testbeds. This transition illustrates the progressive maturation of DECT-2020 NR from theoretical protocol design toward practical deployment.

3.3.4. Validation Approaches

To further characterize the maturity of the published research, the experimental studies were classified according to their validation approaches, as illustrated in Figure 7. Hardware prototypes constitute the most frequently employed validation platform, followed by commercial devices, numerical simulations, and system-level simulations. Link-level simulations, field trials, software-defined radio implementations, and industrial measurements are also represented within the reviewed literature.

The diversity of validation methodologies demonstrates that DECT-2020 NR research has progressed beyond purely theoretical investigations. The increasing availability of commercial chipsets, embedded development platforms, and industrial prototype systems has enabled researchers to evaluate protocol implementations under realistic operating conditions. Although large-scale industrial deployments remain comparatively limited, the growing number of hardware-based validation studies indicates that the technology has entered an important phase of practical implementation and experimental verification.

3.3.5. Reported Performance Metrics

Figure 8 summarizes the performance metrics reported throughout the reviewed literature. Packet error rate, energy consumption, latency, reliability, and bit error rate are the most frequently evaluated performance indicators, emphasizing the importance of communication reliability and robustness in the design objectives of DECT-2020 NR. Additional metrics including throughput, communication coverage, received signal strength, signal-to-noise ratio, fairness, localization accuracy, and connection density are reported less frequently, while resource utilization and spectral efficiency receive comparatively limited attention.

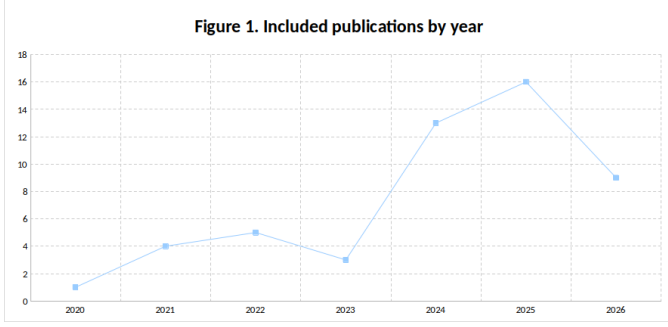


Figure 2: Annual distribution of the included DECT-2020 NR publications.

The observed distribution reflects the current priorities of the research community. Existing studies primarily focus on demonstrating reliable communication, validating protocol correctness, and satisfying the stringent performance requirements associated with industrial wireless communication systems. In contrast, comparatively fewer publications investigate higher-level network optimization metrics such as fairness, resource utilization, and intelligent scheduling. These observations suggest that future research will increasingly address communication efficiency and resource optimization as the technology matures and larger-scale deployments become available.

3.3.6. Implementation Platforms

The implementation platforms employed in the reviewed studies are summarized in Figure 9. Nordic Semiconductor’s nRF91 platform represents the most widely adopted hardware platform for experimental DECT-2020 NR implementations, reflecting its availability as one of the first commercial development platforms supporting the technology. MATLAB remains the dominant software environment for analytical modeling and simulation-based evaluations, while FPGA implementations, GNU Radio, USRP-based software-defined radio platforms, Zephyr, and open-source software frameworks are employed in a smaller number of studies.

The diversity of implementation platforms demonstrates the continued evolution of DECT-2020 NR from algorithmic research toward practical system development. Software simulation environments remain essential for protocol analysis and performance evaluation, whereas commercial embedded platforms increasingly enable real-world experimentation and industrial validation. This combination of software- and hardware-based research provides strong evidence that the technology is transitioning from standardization and theoretical development toward practical deployment in industrial communication systems.

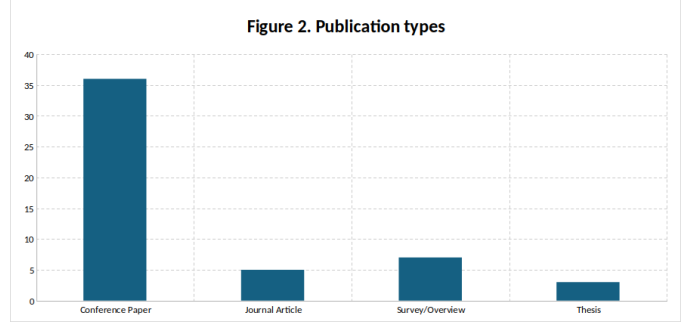


Figure 3: Annual distribution of the included DECT-2020 NR publications.

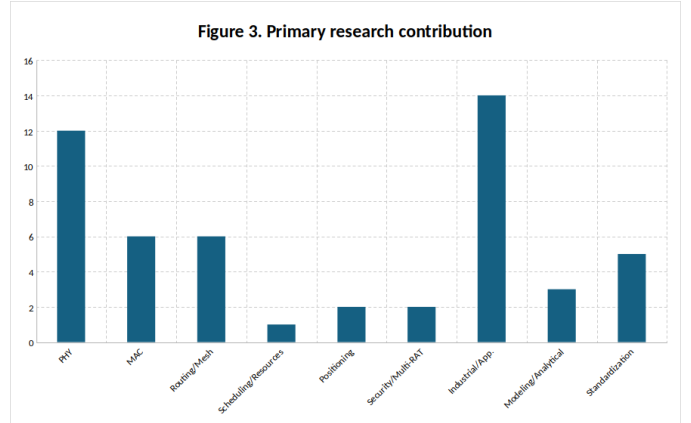


Figure 4: Annual distribution of the included DECT-2020 NR publications.

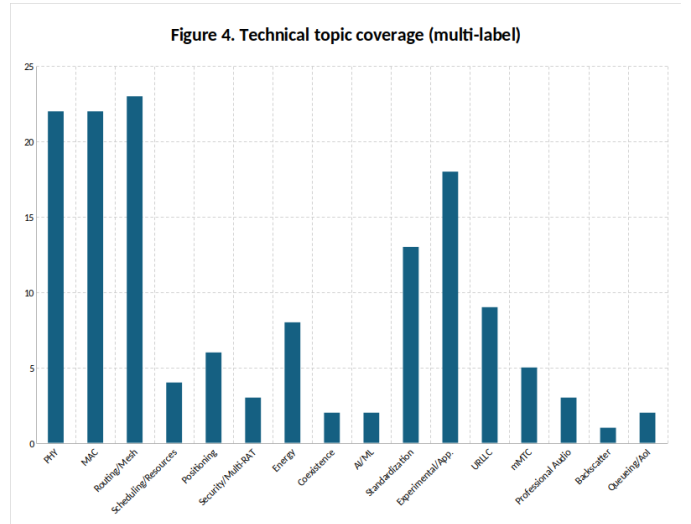


Figure 5: Annual distribution of the included DECT-2020 NR publications.

4. Classification of Existing Research

4.1. Standardization

4.2. System Architecture

4.3. Physical Layer

4.4. MAC Layer

4.5. Resource Allocation and Scheduling

4.6. Implementation Platforms

10 4.7. Performance Evaluation

4.8. Industrial Applications

4.9. Security

5. Comparative Analysis

6. Research Gaps

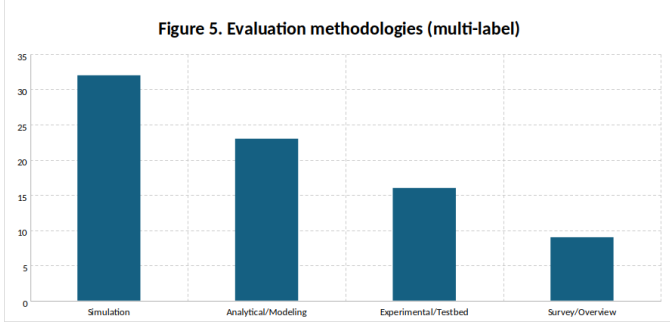


Figure 6: Annual distribution of the included DECT-2020 NR publications.

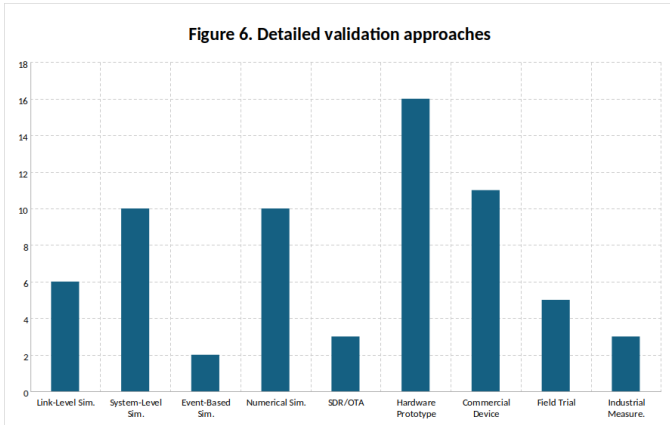


Figure 7: Annual distribution of the included DECT-2020 NR publications.

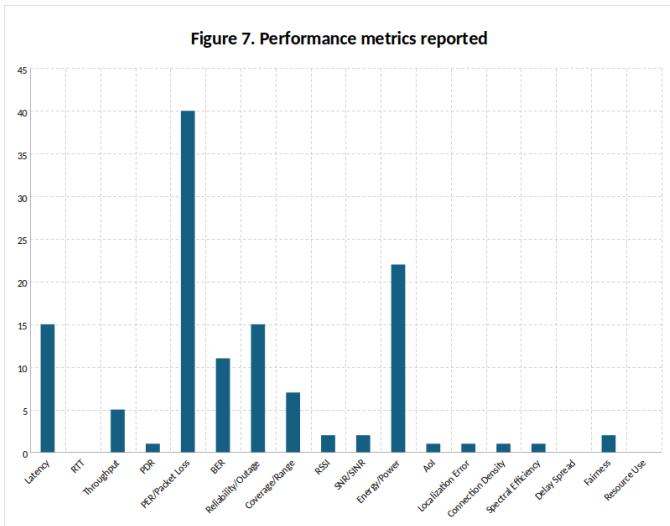


Figure 8: Annual distribution of the included DECT-2020 NR publications.

DECT Digital Enhanced Cordless Telecommunications.

DECT-2020 NR DECT-2020 New Radio.

eMBB Enhanced Mobile Broadband.

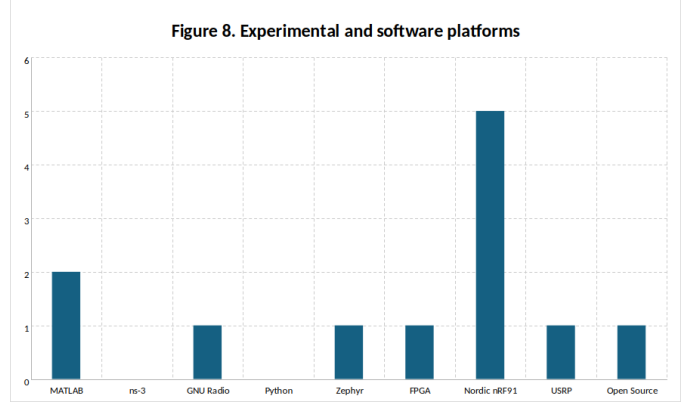


Figure 9: Annual distribution of the included DECT-2020 NR publications.

ETSI European Telecommunications Standards Institute.

FPGA Field-Programmable Gate Array.

IIoT Industrial Internet of Things.

IMT-2020 International Mobile Telecommunications-2020.

ITU-R International Telecommunication Union Radio-communication Sector.

KPI Key Performance Indicator.

MAC Medium Access Control.

mMTC Massive Machine-Type Communications.

PHY Physical Layer.

PRISMA Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

QA Quality Assessment.

RIT Radio Interface Technology.

RQ Research Question.

SLR Systematic Literature Review.

URLLC Ultra-Reliable Low-Latency Communications.

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